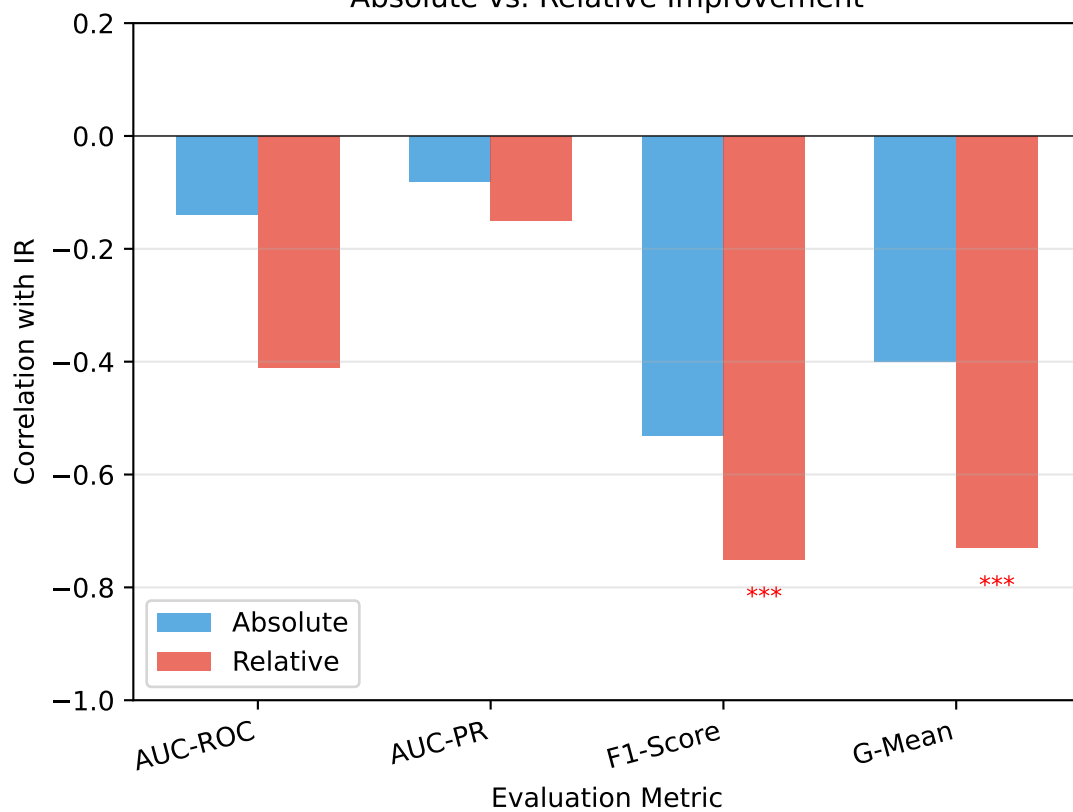
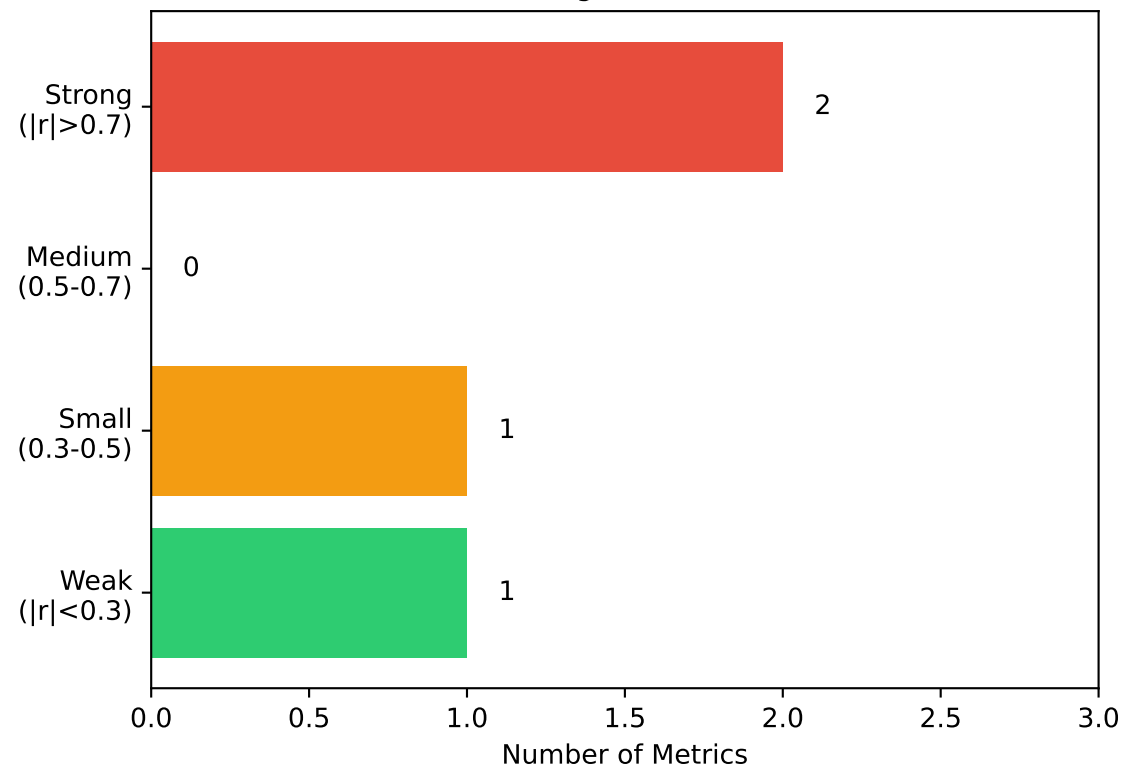


Validation Experiments: Ceiling Effects and Metric-Dependence

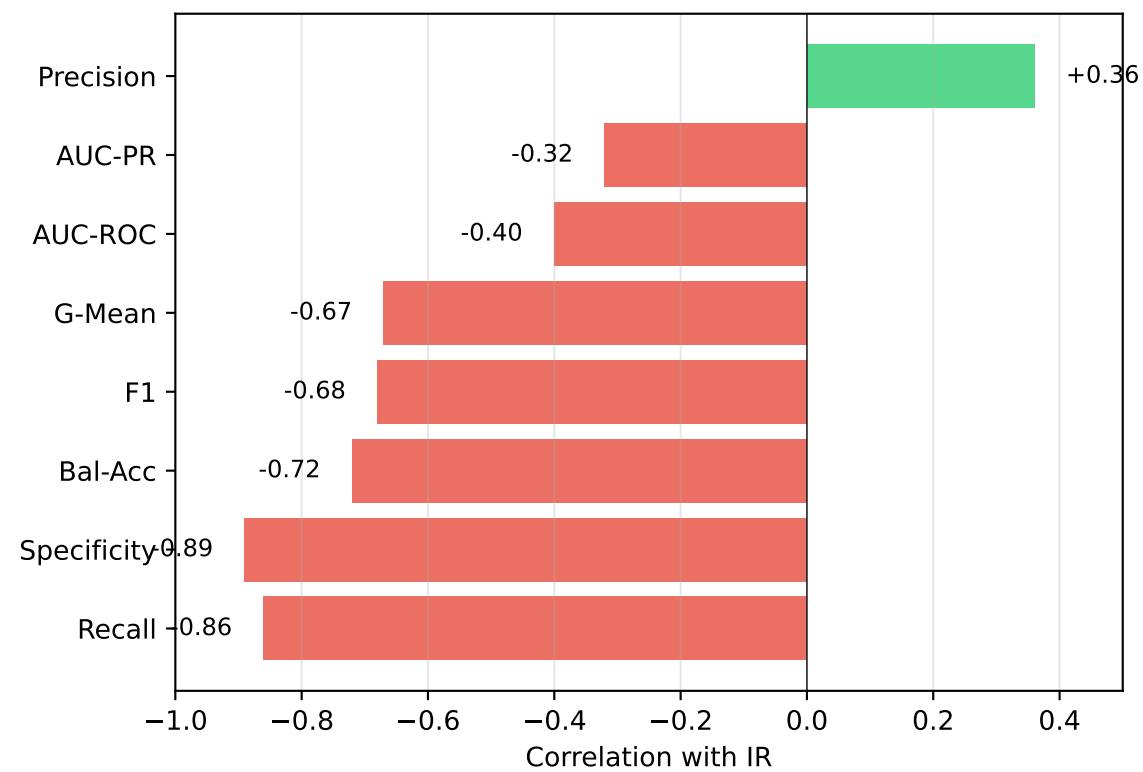
(A) Ceiling Effect Control
Absolute vs. Relative Improvement



(B) Effect Size Distribution
(Ceiling-Controlled)



(C) Multi-Metric Validation
8 Evaluation Metrics



(D) Metric Category Comparison
(Mean $\pm$ SD)

